A7.2 Exclusive accesses

The exclusive access mechanism can provide semaphore-type operations without requiring the bus to remain dedicated to a particular master for the duration of the operation. This means the semaphore-type operations do not impact either the bus access latency or the maximum achievable bandwidth.

The **AxLOCK** signals select exclusive access, and the **RRESP** and **BRESP** signals indicate the success or failure of the exclusive access read or write respectively.

The slave requires additional logic to support exclusive access. The AXI protocol provides a mechanism to indicate when a master attempts an exclusive access to a slave that does not support it. The remainder of this section describes the AXI Exclusive access mechanism.

A7.2.1 Exclusive access process

The basic mechanism of an exclusive access is:

1. A master performs an exclusive read from an address.
2. At some later time, the master attempts to complete the exclusive operation by performing an exclusive write to the same address, and with an **AWID** that matches the **ARID** used for the exclusive read.
3. This exclusive write access is signaled as either:
   - Successful, if no other master has written to that location since the exclusive read access. In this case the exclusive write updates memory.
   - Failed, if another master has written to that location since the exclusive read access. In this case the memory location is not updated.

   A master might not complete the write portion of an exclusive operation. The exclusive access monitoring hardware monitors only one address for each transaction ID. If a master does not complete the write portion of an exclusive operation, a subsequent exclusive read by that master using the same transaction ID changes the address that is being monitored for exclusive accesses.

A7.2.2 Exclusive access from the perspective of the master

A master starts an exclusive operation by performing an exclusive read. If the transaction is successful, the slave returns the EXOKAY response, indicating that the slave recorded the address to be monitored for exclusive accesses.

If the master attempts an exclusive read from a slave that does not support exclusive accesses, the slave returns the OKAY response instead of the EXOKAY response.

Note

The master can treat the OKAY response as an error condition indicating that the exclusive access is not supported. This specification recommends that the master does not perform the write portion of this exclusive operation.

At some time after the exclusive read, the master tries an exclusive write to the same location. If the contents of the addressed location have not changed since the exclusive read, the exclusive write operation succeeds. The slave returns the EXOKAY response, and updates the memory location.

If the contents of the addressed location have changed since the exclusive read, the exclusive write attempt fails, and the slave returns the OKAY response instead of the EXOKAY response. The exclusive write attempt does not update the memory location.

A master might not complete the write portion of an exclusive operation. If this happens, the slave continues to monitor the address for exclusive accesses until another exclusive read starts a new exclusive access sequence.

A master must not start the write part of an exclusive access sequence until the read part is complete.

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A7.2.3 Exclusive access from the perspective of the slave

A slave that does not support exclusive accesses can ignore the **AxLOCK** signals. It must provide an OKAY response for both normal and exclusive accesses.

A slave that supports exclusive access must have monitor hardware. This specification recommends that such a slave has a monitor unit for each exclusive-capable master ID that can access it. The *ARM Architecture Reference Manual, ARMv7-A and ARMv7-R edition* defines an exclusive access monitor, and a single-ported slave can have such an exclusive access monitor external to the slave. A multiported slave might require internal monitoring.

The exclusive access monitor records the address and **ARID** value of any exclusive read operation. Then it monitors that location until either a write occurs to that location or until another exclusive read with the same **ARID** value resets the monitor to a different address.

When the slave receives an exclusive write with a given **AWID** value, the monitor checks to see if that address is being monitored for exclusive access with that **AWID**. If it is, then this indicates that no write has occurred to that location since the exclusive read access, and the exclusive write proceeds, completing the exclusive access. The slave returns the EXOKAY response to the master, and updates the addressed memory location.

If the address is no longer being monitored with the same **AWID** value at the time of an exclusive write, this indicates one of the following:

- the location has been updated since the exclusive read access
- the monitor has been reset to another location.

In both cases the exclusive write must not update the addressed location, and the slave must return the OKAY response instead of the EXOKAY response.

A7.2.4 Exclusive access restrictions

The following restrictions apply to exclusive accesses:

- The burst size and burst length of an exclusive write with a given ID must be the same as the burst size and burst length of the preceding exclusive read with the same ID.
- The address of an exclusive access must be aligned to the total number of bytes in the transaction, that is, the product of the burst size and burst length.
- The addresses for the exclusive read and the exclusive write must be identical.
- The **ARID** value of the exclusive read must match the **AWID** value of the exclusive write.
- The control signals for the exclusive read and exclusive write transactions must be identical.
- The number of bytes to be transferred in an exclusive access burst must be a power of 2, that is, 1, 2, 4, 8, 16, 32, 64, or 128 bytes.
- The maximum number of bytes that can be transferred in an exclusive burst is 128.
- In AXI4, the burst length for an exclusive access must not exceed 16 transfers.
- The value of the **AxCACHE** signals must guarantee that the slave that is monitoring the exclusive access sees the transaction. For example, an exclusive access must not have an **AxCACHE** value that indicates that the transaction is Cacheable.

Failure to observe these restrictions causes UNPREDICTABLE behavior.

The minimum number of bytes to be monitored during an exclusive operation is defined by the burst length and burst size of the transaction. The slave can monitor a larger number of bytes, up to 128 which is the maximum size of an exclusive access. However, this can result in a successful exclusive access being indicated as failing because a neighboring byte was updated.

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A7.2.5 Slaves that do not support exclusive access

The response signals, **RRESP** and **BRESP**, include an OKAY response for successful normal accesses and an EXOKAY response for successful exclusive accesses. This means that a slave that does not support exclusive accesses can provide an OKAY response to indicate the failure of an exclusive access.

Note

- An exclusive write to a slave that does not support exclusive access always updates the memory location.
- An exclusive write to a slave that supports exclusive access updates the memory location only if the exclusive write is successful.

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